

Robust Sparse Portfolio

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Abstract -

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1 Introduction

We consider a market with n risky assets and a riskless asset whose period return is r_c . The risky assets have an unknown mean return vector r and a positive definite covariance matrix D . The initial wealth is normalized to 1. We define the ellipsoidal uncertainty set $U_r = \{r : \|r - \hat{r}\|_D \leq \gamma\}$, where $\gamma > 0$ is a tolerance parameter and $\hat{r}$ denotes the estimated mean return vector; additionally, $\bar{r}$ denotes a given target wealth level. The *robust* mean–variance portfolio choice problem is formulated as

$$\min_{x \in \mathbb{R}^n} x^\top D x \quad \text{s.t.} \quad r^\top x + r_c(1 - \mathbf{1}^\top x) \geq \bar{r}, \quad \forall r \in U_r,$$

and this semi-infinite problem is equivalent to

$$\min_{x \in \mathbb{R}^n} x^\top D x \quad \text{s.t.} \quad \hat{r}^\top x + r_c(1 - \mathbf{1}^\top x) - \gamma\|x\|_D \geq \bar{r}.$$

Letting $\beta > 0$ denote the penalty parameter, $\mathbf{r} = \hat{r} - r_c \mathbf{1}$ the estimated excess mean return vector, and $\bar{\mathbf{r}} = \bar{r} - r_c$, we define the sparse robust mean–variance portfolio choice problem:

$$\min_{x \in \mathbb{R}^n} x^\top D x + \beta\|x\|_0 \quad \text{s.t.} \quad \mathbf{r}^\top x - \gamma\|x\|_D \geq \bar{\mathbf{r}}. \quad (\text{CP-RMVP})$$

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Assumption 1 *Target wealth $\bar{r}$ is strictly greater than r_c the period return of the riskless asset.*

We assume $\bar{r} > r_c$ to exclude the trivial solution $x = \mathbf{0}$. Finally, we denote by $H = \sqrt{\mathbf{r}^\top D^{-1} \mathbf{r}}$ the square root of the optimal Sharpe ratio in the market, corresponding to the slope of the capital market line.

Let $\beta > 0$ denote the sparsity penalty parameter, $\gamma > 0$ a tolerance parameter, and $N \in \mathbb{N}$. Consider $D \in \mathbb{R}^{N \times N}$, $\mathbf{r} \in \mathbb{R}^N$, and $\bar{\mathbf{r}} \in \mathbb{R}$. Define

$$\mathcal{X} = \{x \in \mathbb{R}^N : \mathbf{r}^\top x - \gamma \|x\|_D \geq \bar{\mathbf{r}}\}, \quad (1)$$

$$\mathcal{N}_{\bar{x}} = \{x \in \mathbb{R}^N : \mathbf{r}^\top x \geq \gamma(\|\bar{x} + x\|_D - \|\bar{x}\|_D)\}, \quad \bar{x} \in \mathcal{X}, \quad (2)$$

$$\mathcal{F}_\beta(x) = x^\top D x + \beta \|x\|_0. \quad (3)$$

Here, $\|x\|_0 = |\sigma(x)|$, where $\sigma(x)$ is the support of x (indices of nonzero entries), and $|\cdot|$ denotes cardinality.

We also introduce the indicator $\phi : \mathbb{R} \rightarrow \{0, 1\}$ defined by

$$\phi(t) = \begin{cases} 0, & t = 0, \\ 1, & t \neq 0, \end{cases}$$

so that

$$\|x\|_0 = \sum_{i=1}^N \phi(x[i]) = \sum_{i \in \sigma(x)} \phi(x[i]).$$

With this, (3) can be expressed as

$$\mathcal{F}_\beta(x) = x^\top D x + \beta \sum_{i \in \sigma(x)} \phi(x[i]).$$

Let $\mathbb{I}_N = (\{1, \dots, N\}, <)$ be the strictly ordered index set, where $<$ denotes the standard order. Any subset $\omega \subseteq \mathbb{I}_N$ inherits this property. We denote the all-ones vector by $\mathbf{1}$, the identity matrix by $\mathbf{I}$, and the zero vector by $\mathbf{0}$. For $x \in \mathbb{R}^N$, the ℓ_p norm is defined for $1 \leq p \leq \infty$ as

$$\|x\|_p := \left(\sum_{i=1}^N |x[i]|^p \right)^{1/p}.$$

For a matrix $A \in \mathbb{R}^{M \times N}$, the spectral norm is

$$\|A\|_2 = \max_{i=1, \dots, r} \sigma_i(A),$$

where $\sigma_i(A)$ is the i -th singular value and $r = \min\{M, N\}$. More generally, the operator norm from ℓ_p to ℓ_q is

$$\|A\|_{p,q} := \sup_{\|x\|_p \leq 1} \|Ax\|_q.$$

Given $x \in \mathbb{R}^N$ and $\rho > 0$, the open ℓ_p ball of radius ρ centered at x is

$$B_p(x, \rho) := \{y \in \mathbb{R}^N : \|x - y\|_p < \rho\}.$$

We denote the i -th column of D by d_i . For $\omega \subseteq \mathbb{I}_N$, the following notation for subvectors and submatrices will be used:

$$r_\omega := (r[\omega[1]], \dots, r[\omega[|\omega|]]) \in \mathbb{R}^{|\omega|},$$

$$D_\omega := ((d_{\omega[1]})_\omega, \dots, (d_{\omega[|\omega|]})_\omega) \in \mathbb{R}^{|\omega| \times |\omega|}.$$

We also define the zero-padding operator $Z_\omega : \mathbb{R}^{|\omega|} \rightarrow \mathbb{R}^N$ by

$$x = Z_\omega(x_\omega), \quad x[i] = \begin{cases} 0, & i \notin \omega, \\ x_\omega[k], & \text{if } \omega[k] = i. \end{cases}$$

Note that for $x \in \mathbb{R}^N$ and $\omega \supseteq \sigma(x)$,

$$x^T D x = x_\omega^T D_\omega x_\omega.$$

This property motivates searching for local minimizers by restricting to supports.

Finally, we impose the following assumption.

Assumption 2 *The matrix D is positive definite, and the tolerance level of uncertainty γ satisfies,*

$$\min_{i \in \mathbb{I}_N} \frac{|\mathbf{r}[i]|}{\sqrt{d_i[i]}} > \gamma,$$

where $\sqrt{d_i[i]}$ denotes the square root of i -th diagonal entry of D .

For $\omega \subseteq \mathbb{I}_N$, define

$$K_\omega = \{x \in \mathbb{R}^N : x[i] = 0, \forall i \in \omega^c\}.$$

We study the following subproblem to characterize local minimizers of (RMVP):

$$\min_{x \in \mathcal{X} \cap K_\omega} x^T D x. \tag{RMVP}_\omega$$

Since D is positive definite, the objective is strictly convex, and hence the problem has a unique solution whenever feasible.

We further define

$$\begin{aligned}\mathcal{X}_\omega &= \left\{ u \in \mathbb{R}^{|\omega|} : \mathbf{r}_\omega^T u - \gamma \|u\|_{D_\omega} \geq \bar{\mathbf{r}} \right\}, \\ (\mathcal{N}_\omega)_{\bar{u}} &= \left\{ u \in \mathbb{R}^{|\omega|} : \mathbf{r}_\omega^T u \geq \gamma (\|\bar{u} + u\|_{D_\omega} - \|\bar{u}\|_{D_\omega}) \right\}.\end{aligned}$$

Using these sets and the zero-padding operator, we define the equivalent problem

$$\min_{u \in \mathcal{X}_\omega} u^T D_\omega u, \quad |\omega| \geq 1. \quad (\text{ZP-RMVP}_\omega)$$

Remark 1 Assumption 2 extends to all subproblems: for any $\omega \subseteq \mathbb{I}_N$, D_ω is positive definite and $\mathbf{r}_\omega[i] > \gamma(\sqrt{d_\omega})_i[i]$ holds for all $i \in \mathbb{I}_{|\omega|}$.

Proposition 1 *Let $\beta > 0$ and $\omega \subseteq \mathbb{I}_N$. The unique optimal solution of (ZP-RMVP_ω) is*

$$\hat{u} = \left(\frac{\bar{\mathbf{r}}}{H_\omega(H_\omega - \gamma)} \right) (D_\omega)^{-1} \mathbf{r}_\omega,$$

where $H_\omega = \sqrt{\mathbf{r}_\omega^T (D_\omega)^{-1} \mathbf{r}_\omega}$.

Proof The proof parallels the argument in ROBUSTPORTMCP.

For the branching algorithm, we require a closed-form expression for the dual variable associated with the feasibility constraint.

Proposition 2 *Let $\beta > 0$ and $\omega \subseteq \mathbb{I}_N$. If $\hat{u} \in \mathbb{R}^{|\omega|}$ is an optimal solution of (ZP-RMVP_ω) , then the optimal Lagrange multiplier μ^* associated with the constraint in (ZP-RMVP_ω) is given by*

$$\mu^* = \frac{2\bar{\mathbf{r}}}{(H_\omega - \gamma)^2}.$$

Moreover, the restricted constraint is active at optimality, i.e. $\mathbf{r}_\omega^T \hat{u} - \gamma \|\hat{u}\|_{D_\omega} = \bar{\mathbf{r}}$.

Proof The proof parallels the argument in ROBUSTPORTMCP.

We also verify the activity of the restricted constraint. If the constraint was not active, the complementary slackness condition would imply $\mu = 0$. However, Assumptions 1 and 2 imply that $H_\omega \neq \gamma$ and $\bar{\mathbf{r}} > 0$, which creates a contradiction. Hence, the optimal solution $\hat{u}$ of (ZP-RMVP_ω) satisfies $\mathbf{r}_\omega^T \hat{u} - \gamma \|\hat{u}\|_{D_\omega} = \bar{\mathbf{r}}$. $\square$

Lemma 1 (Fenchel Completion to a Square) *For any $v \in \mathbb{R}^N$, we have*

$$v^T D^{-1} v = \max_{u \in \mathbb{R}^N} (2u^T v - u^T D u).$$

Proof Let us fix $v \in \mathbb{R}^N$ and $u \in \mathbb{R}^N$ be arbitrary. We define $Q(u) = 2u^T v - u^T Dv$. Since D is positive definite we have:

$$Q(u) = -(-2u^T v + u^T Dv) = -\underbrace{(-2u^T v + u^T Dv + v^T A^{-1}v)}_{*} + v^T D^{-1}v.$$

Since $(*)$ is always non-positive, maximum is attained at $u = D^{-1}v$. Hence, we the desired identity. $\square$

Remark 2 Let $u \in \mathcal{X}_\omega$, and write $u = tv$ with $t > 0$ and $\|v\|_{D_\omega} = 1$. Then

$$t(\gamma - \mathbf{r}_\omega^T v) + \bar{\mathbf{r}} \leq 0.$$

Since $\bar{\mathbf{r}} > 0$, feasibility requires $\gamma - \mathbf{r}_\omega^T v < 0$. This holds whenever

$$\max_{\|v\|_{D_\omega}=1} \mathbf{r}_\omega^T v = \sqrt{\mathbf{r}_\omega^T (D_\omega)^{-1} \mathbf{r}_\omega} =: H_\omega > \gamma.$$

To guarantee feasibility across all supports, it suffices to ensure

$$\min_{\omega \neq \emptyset} H_\omega > \gamma.$$

For any $\omega \neq \emptyset$ and $i \in \omega$ we use Lemma 1 by taking $u = te_i \in \mathbb{R}^{|\omega|}$ and obtain:

$$H_\omega = \sqrt{\mathbf{r}_\omega^T (D_\omega)^{-1} \mathbf{r}_\omega} \geq \sqrt{2t\mathbf{r}_\omega[i] - t^2 d_i[i]}.$$

Maximum on the right hand side is attained at $t = \mathbf{r}_\omega[i]^2 / d_i[i]$ and we have

$$\sqrt{\mathbf{r}_\omega^T (D_\omega)^{-1} \mathbf{r}_\omega} \geq \frac{\mathbf{r}_\omega[i]}{\sqrt{d_i[i]}} \Rightarrow \sqrt{\mathbf{r}_\omega^T (D_\omega)^{-1} \mathbf{r}_\omega} \geq \max_{i \in \omega} \frac{|\mathbf{r}_\omega[i]|}{\sqrt{d_i[i]}}.$$

In fact, the minimum is attained on singletons:

$$\min_{\omega \neq \emptyset} H_\omega = H_{\min} := \min_{i \in \mathbb{I}_N} \frac{|\mathbf{r}[i]|}{\sqrt{d_i[i]}} > \gamma.$$

Assumption 2 provides this bound. Thus, strict feasibility and convexity ensure uniqueness of the solution.

Remark 3 For convenience, define for $\omega \subseteq \mathbb{I}_N$, $|\omega| \geq 1$,

$$\xi(\omega) = \left(\frac{\bar{\mathbf{r}}}{H_\omega(H_\omega - \gamma)} \right) (D_\omega)^{-1} \mathbf{r}_\omega, \quad \Xi(\omega) = Z_\omega(\xi(\omega)).$$

Lemma 2 *Problems (ZP-RMVP $_\omega$) and (RMVP $_\omega$) are equivalent.*

Proof The map $Z_\omega : \mathcal{X}_\omega \rightarrow \mathcal{X} \cap K_\omega$ is a bijection. Moreover, for any $x_\omega \in \mathcal{X}_\omega$,

$$x_\omega^T D_\omega x_\omega = Z_\omega(x_\omega)^T D Z_\omega(x_\omega).$$

Hence the two formulations are equivalent. $\square$

Remark 4 For $\omega \subseteq \mathbb{I}_N$ with $|\omega| \geq 1$, the point $\Xi(\omega) \in \mathbb{R}^N$ is the unique solution of (RMVP_ω) .

Proposition 3 Let $\beta > 0$ and $\hat{x} \in \mathcal{X}$. Define $\hat{\sigma} = \sigma(\hat{x})$ and

$$\rho := \min \left\{ \min_{i \in \hat{\sigma}} |\hat{x}[i]|, \frac{\beta}{2(\|D\hat{x}\|_1 + 1)} \right\}.$$

Then $\rho > 0$, and:

(i) If $y \in B_\infty(0, \rho)$, then

$$\sum_{i \in \mathbb{I}_N} \phi(\hat{x}[i] + y[i]) = \sum_{i \in \hat{\sigma}} \phi(\hat{x}[i]) + \sum_{i \in \hat{\sigma}^c} \phi(y[i]).$$

(ii) If $y \in B_\infty(0, \rho) \cap (\mathbb{R}^N \setminus K_{\hat{\sigma}})$, then

$$\mathcal{F}_\beta(\hat{x} + y) \geq \mathcal{F}_\beta(\hat{x}),$$

with strict inequality whenever $\hat{\sigma}^c \neq \emptyset$.

Proof The proof follows identically from the argument in SPARSEPORTBUSE.

Remark 5 For $\hat{x} \in \mathcal{X}$ and $\beta > 0$, Proposition 8 remains valid if perturbations are restricted to $\mathcal{N}_{\hat{x}}$, ensuring the feasibility of $\hat{x} + y$.

Proposition 4 Let $\omega \subseteq \mathbb{I}_N$, $\omega \neq \emptyset$. For any $\beta > 0$, the objective $\mathcal{F}_\beta$ reaches a (local) minimum of (CP-RMVP) at $\Xi(\omega)$, with

$$|\sigma(\Xi(\omega))| \geq 1, \quad \sigma(\Xi(\omega)) \subseteq \omega.$$

Proof The proof parallels the argument in SPARSEPORTBUSE.

Lemma 3 Let $\beta > 0$ and let $\hat{x}$ be a (local) minimizer of (CP-RMVP). Then

$$\hat{x} = \Xi(\sigma(\hat{x})).$$

Proof The proof again follows directly from SPARSEPORTBUSE. $\square$

Proposition 5 For $\beta > 0$ and let (CP-RMVP) have a global minimum at $\hat{x}$. Let $\hat{\sigma} = \sigma(\hat{x})$. Then

$$\|\hat{x}\|_\infty \leq \sqrt{\frac{\eta}{\lambda_*(D)}} < \infty,$$

where $\eta = \min_{i \in \mathbb{I}_N} \left\{ \frac{\bar{\tau}^2 d_i[i]}{(\mathfrak{r}[i] - \gamma \sqrt{d_i[i]})^2} \right\}$ and $\lambda_*(D)$ is the smallest eigenvalue of the matrix D .

Proof Let $\hat{x}$ be a global minimizer of CP-RMVP. Then, for any $i \in \mathbb{I}_N$, we have

$$\hat{x}^T D \hat{x} + \beta \|\hat{x}\|_0 \leq \Xi(\{i\})^T D \Xi(\{i\}) + \beta \Rightarrow \|\hat{x}\|_D^2 \leq \min_{i \in \mathbb{I}_N} \Xi(\{i\})^T D \Xi(\{i\}) = \min_{i \in \mathbb{I}_N} \left\{ \frac{\bar{\tau}^2 d_i[i]}{(\mathfrak{r}[i] - \gamma \sqrt{d_i[i]})^2} \right\}.$$

This implies that $\|\hat{x}\|_D \leq \sqrt{\eta}$. For an upper bound on the maximal entry, it is sufficient to rescale η with the smallest eigenvalue, that is $\|\hat{x}\|_\infty \leq \sqrt{\frac{\eta}{\lambda_*(D)}}$. $\square$

Theorem 3 For $\beta > 0$ and let (CP-RMVP) have a global minimum at $\hat{x}$. Let $\hat{\sigma} = \sigma(\hat{x})$ and for all $i \in \mathbb{I}_N$ we define

$$\rho_i = \max_{j \neq i} \left\{ \left\| e_i \pm \frac{|\mathbf{r}[i]| + \gamma\sqrt{d_i[i]}}{|\mathbf{r}[j]| - \gamma\sqrt{d_j[j]}} e_j \right\|_D \right\} \text{ and } \eta = \min_{i \in \mathbb{I}_N} \left\{ \frac{\bar{\mathbf{r}}^2 d_i[i]}{(\mathbf{r}[i] - \gamma\sqrt{d_i[i]})^2} \right\}.$$

Then

$$i \in \hat{\sigma} \Rightarrow |\hat{x}[i]| \geq \min \left\{ \frac{\sqrt{\eta + \beta} - \sqrt{\eta}}{\rho_i}, \frac{\bar{\mathbf{r}}}{|\mathbf{r}[i]| - \gamma\sqrt{d_i[i]}} \right\}.$$

Proof Let $i \in \hat{\sigma}$. Assume $|\hat{\sigma}| \geq 2$, then there exists $j \in \hat{\sigma}$ such that $i \neq j$. We define $g_{ij} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ as

$$g_{ij}(x) = x - x[i]e_i - x[j]e_j$$

where e_i and e_j are canonical basis vectors of the specified index. We introduce the function

$$f(t_i, t_j) = \mathcal{F}_\beta(g_{ij}(\hat{x}) + t_i e_i + t_j e_j).$$

Finally, we define a feasibility function

$$h(t_i, t_j) = \mathbf{r}^T g_{ij}(\hat{x}) + t_i \mathbf{r}[i] + t_j \mathbf{r}[j] - \gamma \|g_{ij}(\hat{x}) + t_i e_i + t_j e_j\|_D - \bar{\mathbf{r}}.$$

Since $\hat{x}$ is a global minimizer, it should be a minimizer of $(RMVP_{\hat{\sigma}})$ and in Proposition 2 we have shown that the restricted constraint is satisfied actively at $\hat{x}$. But, this also implies the main constraint is satisfied actively and $h(\hat{x}[i], \hat{x}[j]) = 0$. Precisely, we rewrite this equality as

$$h(\hat{x}[i], \hat{x}[j]) = \mathbf{r}^T \hat{x} - \gamma \|\hat{x}\|_D - \bar{\mathbf{r}} = 0 \Rightarrow \bar{\mathbf{r}} = \mathbf{r}^T \hat{x} - \gamma \|\hat{x}\|_D.$$

Now, we define a map by fixing a choice for t_j

$$\begin{aligned} h(t_i, t_j) &= \mathbf{r}^T g_{ij}(\hat{x}) + t_i \mathbf{r}[i] + t_j \mathbf{r}[j] - \gamma \|g_{ij}(\hat{x}) + t_i e_i + t_j e_j\|_D - \bar{\mathbf{r}} \\ &= \gamma (\|\hat{x}\|_D - \|g_{ij}(\hat{x}) + t_i e_i + t_j e_j\|_D) + (t_i - \hat{x}[i])\mathbf{r}[i] + (t_j - \hat{x}[j])\mathbf{r}[j] \\ &\geq -\gamma (\|(t_i - \hat{x}[i])e_i + (t_j - \hat{x}[j])e_j\|_D) + (t_i - \hat{x}[i])\mathbf{r}[i] + (t_j - \hat{x}[j])\mathbf{r}[j] \\ &\geq -\gamma |t_i - \hat{x}[i]| \sqrt{d_i[i]} - \gamma |t_j - \hat{x}[j]| \sqrt{d_j[j]} + (t_i - \hat{x}[i])\mathbf{r}[i] + (t_j - \hat{x}[j])\mathbf{r}[j] \end{aligned}$$

Now assume that $t_j \geq \hat{x}[j]$. If the resulting lower bound on t_j satisfies the assumption we may omit it.

$$h(t_i, t_j) \geq -\gamma |t_i - \hat{x}[i]| \sqrt{d_i[i]} + (t_i - \hat{x}[i])\mathfrak{r}[i] + (t_j - \hat{x}[j])(\mathfrak{r}[j] - \gamma \sqrt{d_j[j]})$$

Now if we set

$$t_j = |t_i - \hat{x}[i]| \frac{|\mathfrak{r}[i]| + \gamma \sqrt{d_i[i]}}{|\mathfrak{r}[j]| - \gamma \sqrt{d_j[j]}} + \hat{x}[j] = |t_i - \hat{x}[i]| \delta_{ij} + \hat{x}[j]$$

we will have $\delta_{ij} > 0$, $t_j \geq \hat{x}[j]$ and $h(t_i, t_j) \geq 0$. Hence, we may define a 1-dimensional restricted objective as

$$\begin{aligned} f(t) &= \|g_{ij}(\hat{x}) + te_i + (|t - \hat{x}[i]| \delta_{ij} + \hat{x}[j])e_j\|_D^2 + \beta \|g_{ij}(\hat{x}) + te_i + (|t - \hat{x}[i]| \delta_{ij} + \hat{x}[j])e_j\|_0 \\ &= \|\hat{x} + (t - \hat{x}[i])e_i + |t - \hat{x}[i]| \delta_{ij}e_j\|_D^2 + \beta \|\hat{x} + (t - \hat{x}[i])e_i + |t - \hat{x}[i]| \delta_{ij}e_j\|_0 \end{aligned}$$

Optimality of $\hat{x}[i]$ implies, $f(0) \geq f(t)$, hence we have

$$\|\hat{x}\|_D^2 + \beta \|\hat{x}\| \leq \|\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j\|_D^2 + \beta \|\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j\|_0$$

Now, we know $\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j$ has at most $|\hat{\sigma}| - 1$ non-zero elements and this implies the following inequality

$$\begin{aligned} \beta &\leq \|\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j\|_D^2 - \|\hat{x}\|_D^2 = (2\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j)^T D (-\hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j) \\ &\leq \|2\hat{x} - \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j\|_D \|- \hat{x}[i]e_i + |\hat{x}[i]| \delta_{ij}e_j\|_D \\ &= (2\|\hat{x}\|_D + |\hat{x}[i]| \|e_i - \text{sign}(\hat{x}[i])\delta_{ij}e_j\|_D) |\hat{x}[i]| \|e_i - \text{sign}(\hat{x}[i])\delta_{ij}e_j\|_D. \end{aligned} \tag{4}$$

From Proposition 5 we have $\|\hat{x}\|_D \leq \sqrt{\eta}$. When we plug the appropriate upper bounds back in 4 we get

$$\beta \leq (2\sqrt{\eta} + |\hat{x}[i]| \rho_i) |\hat{x}[i]| \rho_i. \tag{5}$$

We consider the following polynomial in v

$$P(v) = \rho_i^2 v^2 + 2\sqrt{\eta} \rho_i v - \beta.$$

This polynomial is a quadratic with positive leading coefficient. Hence, for $|\hat{x}[i]|$ to satisfy the inequality in 5, it has to be larger than the positive root of polynomial $P(v)$. Roots of $P(v)$ are given as

$$v_+, v_- = \frac{-2\sqrt{\eta} \rho_i \pm \sqrt{4\eta \rho_i^2 + 4\rho_i^2 \beta}}{2\rho_i^2} = \frac{-\sqrt{\eta} \pm \sqrt{\eta + \beta}}{\rho_i}.$$

Hence, we get the lower bound on $|\hat{x}[i]|$ as

$$|\hat{x}[i]| \geq \frac{\sqrt{\eta + \beta} - \sqrt{\eta}}{\rho_i}.$$

Finally, if $|\hat{\sigma}| = 1$ then we have closed form solutions for the specific entry $i \in \hat{\sigma}$

$$\hat{x}[i] = \frac{\bar{\mathbf{r}}\text{sign}(\mathbf{r}[i])}{|\mathbf{r}[i]| - \gamma\sqrt{d_i[i]}} \Rightarrow |\hat{x}[i]| = \frac{\bar{\mathbf{r}}}{|\mathbf{r}[i]| - \gamma\sqrt{d_i[i]}}.$$

Combining these two cases will result in

$$i \in \hat{\sigma} \Rightarrow |\hat{x}[i]| \geq \min \left\{ \frac{\sqrt{\eta + \beta} - \sqrt{\eta}}{\rho_i}, \frac{\bar{\mathbf{r}}}{|\mathbf{r}[i]| - \gamma\sqrt{d_i[i]}} \right\}.$$

□

ASYMPTOTES? check Auslender...

An alternative robust model is

$$\max_{x \in \mathbb{R}^N} \min_{r \in U_r} r^T x + (1 - \mathbf{1}^T x) r_c \text{ s.t. } x^T D x \leq t^2,$$

for an appropriately chosen $t > 0$. We rewrite the equivalent model

$$\max_{x \in \mathbb{R}^N} \hat{r}^T x + (1 - \mathbf{1}^T x) - \gamma \|x\|_D \text{ s.t. } x^T D x \leq t^2.$$

Hence we rewrite this alternative as

$$\max_{x \in \mathbb{R}^N} \mathbf{r}^T x - \bar{\mathbf{r}} - \gamma \|x\|_D \text{ s.t. } \|x\|_D \leq t.$$

Assumption 4 We assume a lower bound on the tolerance parameter t ,

$$t > \min_{i \in \mathbb{I}_N} \left\{ \frac{\beta}{\frac{|\mathbf{r}[i]|}{\sqrt{d_i[i]}} - \gamma} \right\}.$$

Remark 6 The purpose of this lower bound is to rule out the trivial solution $x = \mathbf{0}$. Let $i \in \mathbb{I}_N$. Then we look for a positive optimal value for the following optimization problem

$$\max_{\|\alpha e_i\|_D \leq t} \mathbf{r}[i] \alpha - \gamma |\alpha| \sqrt{d_i[i]} - \beta = \frac{t}{\sqrt{d_i[i]}} (|\mathbf{r}[i]| - \gamma \sqrt{d_i[i]}) - \beta \Rightarrow t > \frac{\beta}{\frac{|\mathbf{r}[i]|}{\sqrt{d_i[i]}} - \gamma}.$$

It suffices to have this lower bound on t for a single index, hence we take the minimum. This contradicts the optimality of the trivial solution, zero vector.

We define the sparsity regularized form of this alternative in a consistent way for a given $\beta > 0$

$$\min_{x \in \mathbb{R}^N} \mathcal{F}'_\beta(x) := \gamma \|x\|_D - \mathbf{r}^T x + \beta \|x\|_0 \text{ s.t. } \|x\|_D \leq t. \quad (\text{CP-RMVP2})$$

We define the subproblem for $\omega \subseteq \mathbb{I}_N$ as

$$\min_{y \in \mathbb{R}^{|\omega|}} \gamma \|y\|_{D_\omega} - \mathbf{r}_\omega^T y \text{ s.t. } \|y\|_{D_\omega} \leq t. \quad (\text{ZP-RMVP2})$$

Proposition 6 Let $\beta > 0$ and $\omega \subseteq \mathbb{I}_N$. The unique optimal solution of (ZP-RMVP2) is

$$\hat{u} = \frac{t}{H_\omega} (D_\omega)^{-1} \mathbf{r}_\omega.$$

Proof The proof parallels the argument in ROBUSTPORTMCP.

If $H_\omega < \gamma$ then investor only uses the riskless asset. We need to assume that $\min_{\omega \neq \emptyset} H_\omega \geq \gamma$, in other words it suffices to relax the previous assumption or use the same.

Remark 7 For convenience, define for $\omega \subseteq \mathbb{I}_N$, $|\omega| \geq 1$,

$$\xi'(\omega) = \frac{t}{H_\omega} (D_\omega)^{-1} \mathbf{r}_\omega, \quad \Xi'(\omega) = Z_\omega(\xi'(\omega)).$$

Proposition 7 Let $\beta > 0$ and $\omega \subseteq \mathbb{I}_N$. If $\hat{u} \in \mathbb{R}^{|\omega|}$ is an optimal solution of (ZP-RMVP2), then the optimal Lagrange multiplier μ^* associated with the constraint in (ZP-RMVP2) is given by

$$\mu^* = H_\omega - \gamma.$$

Moreover, the restricted constraint is active at optimality, i.e. $\|\hat{u}\|_D = t$.

Proof The proof parallels the argument in ROBUSTPORTMCP.

We also verify the activity of the restricted constraint. If the constraint was not active, the complementary slackness condition would imply $\mu = 0$. However, Assumption 2 implies that $H_\omega > \gamma$, which creates a contradiction. Hence, the optimal solution $\hat{u}$ of (ZP-RMVP2) satisfies $\|\hat{u}\|_D = t$. $\square$

Proposition 8 Let $\beta > 0$ and $\hat{x} \in \overline{B}_D(\hat{x}, t)$. Define $\hat{\sigma} = \sigma(\hat{x})$ and

$$\rho := \min \left\{ \min_{i \in \hat{\sigma}} |\hat{x}[i]|, \frac{\beta}{\|\mathbf{r}\|_1 + \gamma \sqrt{N\lambda^*(D)} + 1} \right\}.$$

where $\lambda^*(D)$ is the largest eigenvalue of the matrix D . Then $\rho > 0$, and:

(i) If $y \in B_\infty(0, \rho)$, then

$$\sum_{i \in \mathbb{I}_N} \phi(\hat{x}[i] + y[i]) = \sum_{i \in \hat{\sigma}} \phi(\hat{x}[i]) + \sum_{i \in \hat{\sigma}^c} \phi(y[i]).$$

(ii) If $y \in B_\infty(0, \rho) \cap (\mathbb{R}^N \setminus K_{\hat{\sigma}})$, then

$$\mathcal{F}_\beta(\hat{x} + y) \geq \mathcal{F}_\beta(\hat{x}),$$

with strict inequality whenever $\hat{\sigma}^c \neq \emptyset$.

Proof (i) For $y \in B_\infty(0, \rho)$ we have $\|y\|_\infty < \min_{i \in \hat{\sigma}} |\hat{x}[i]|$. Then this implies $\phi(\hat{x}[i] + y[i]) = \phi(\hat{x}[i])$ for $i \in \hat{\sigma}$. For $i \in \hat{\sigma}^c$ we have $\phi(\hat{x}[i] + y[i]) = \phi(y[i])$ and it gives the desired result.

Let $y \in B_\infty(0, \rho) \setminus K_{\hat{\sigma}}$. Then

$$\begin{aligned} \mathcal{F}_\beta(\hat{x} + y) &= \gamma \|\hat{x} + y\|_D - \mathbf{r}^T(\hat{x} + y) + \beta \|\hat{x} + y\|_0 \\ &= \mathcal{F}_\beta(\hat{x}) - \mathbf{r}^T y + \gamma(\|\hat{x} + y\|_D - \|\hat{x}\|_D) + \beta \sum_{i \in \hat{\sigma}^c} \phi(y[i]) \\ &\geq \mathcal{F}_\beta(\hat{x}) - \mathbf{r}^T y - \gamma \|y\|_D + \beta \|y_{\hat{\sigma}^c}\|_0 \\ &\geq \mathcal{F}_\beta(\hat{x}) - \|y\|_\infty (\|\mathbf{r}\|_1 + \gamma \sqrt{N\lambda^*(D)}) + \beta \|y_{\hat{\sigma}^c}\|_0. \end{aligned}$$

For $\sigma^c = \emptyset$, inequality is trivial. If not, $\|y_{\hat{\sigma}^c}\|_0 \geq 1$, and the given radius provides the inequality. $\square$

Proposition 9 *Let $\omega \subseteq \mathbb{I}_N$, $\omega \neq \emptyset$. For any $\beta > 0$, the objective $\mathcal{F}_\beta$ reaches a (local) minimum of (CP-RMVP2) at $\Xi'(\omega)$, with*

$$|\sigma(\Xi'(\omega))| \geq 1, \quad \sigma(\Xi'(\omega)) \subseteq \omega.$$

Lemma 4 *Let $\beta > 0$ and let $\hat{x}$ be a (local) minimizer of (CP-RMVP2). Then*

$$\hat{x} = \Xi'(\sigma(\hat{x})).$$

Proof The proof again follows directly from SPARSEPORTBUSE. $\square$

Remark 8 For $\beta > 0$ and let $\hat{x}$ be a global minimizer of CP-RMVP2. In Proposition 5 we developed a non-trivial upper bound on the non-zero entries of a global minimizer. However, for CP-RMVP2 we have a compact feasible region defined by a closed norm ball. Hence, $\|\hat{x}\|_D = t$ and $\|\hat{x}\|_\infty \leq \frac{t}{\sqrt{\lambda_*(D)}}$.

Proposition 10 *For $\beta > 0$ and let (CP-RMVP2) have a global minimum at $\hat{x}$. Let $\hat{\sigma} = \sigma(\hat{x})$. Then*

$$i \in \hat{\sigma} \Rightarrow |\hat{x}[i]| \geq \min \left\{ \frac{\beta}{|\mathbf{r}[i]| + H\sqrt{d_i[i]}}, \frac{t}{\sqrt{d_i[i]}} \right\}.$$

Proof Let $|\hat{\sigma}| \geq 2$ and $i \in \hat{\sigma}$. We define $g_i : \mathbb{R}^N \rightarrow \mathbb{R}^N$ as

$$g_i(x) = x - x[i]e_i$$

where e_i is the canonical basis vector of the specified index. We introduce the function

$$f(t_i) = \mathcal{F}_\beta(g_i(\hat{x}) + t_i e_i).$$

Finally, we define a feasibility function

$$h(t_i) = t - \|g_i(\hat{x}) + t_i e_i\|_D.$$

Since $\hat{x}$ is a global minimizer, it should be a minimizer of $(RMVP2_{\hat{\sigma}})$ and in Proposition 12 we have shown that the restricted constraint is satisfied actively at $\hat{x}$. But, this also implies the main constraint is satisfied actively and $h(\hat{x}[i]) = 0$. Then, we have the following three cases:

Case I: $h(0) \geq 0$, in this case we have $\|g_i(\hat{x})\|_D \leq t$ and due to the global optimality of $\hat{x}$ we have

$$\gamma \|g_i(\hat{x})\|_D - \mathbf{r}^T g_i(\hat{x}) + \beta \|g_{ij}(\hat{x})\|_0 \geq \gamma \|\hat{x}\|_D - \mathbf{r}^T \hat{x} + \beta \|\hat{x}\|_0.$$

These together imply $|\hat{x}[i]| \geq \frac{\beta}{|\mathbf{r}[i]|}$.

Case II: $h(0) < 0$ and $f(0) \geq f(\hat{x})$. Then, similarly we have $|\hat{x}[i]| \geq \frac{\beta}{\gamma\sqrt{d_i[i]} + |\mathbf{r}[i]|}$.

Case III: $f(0) < f(\hat{x})$ and $h(0) < 0$. Since $g_i(\hat{x}) \neq 0$ we can define $\hat{u} = \frac{t}{\|g_i(\hat{x})\|_D} g_i(\hat{x})$. $\hat{u}$ is a feasible point so we require $\mathcal{F}_\beta(\hat{x}) \leq \mathcal{F}_\beta(\hat{u})$. Then we have

$$\gamma t - \mathbf{r}^T \hat{x} + \beta \|\hat{x}\|_0 \leq \gamma t - \mathbf{r}^T \hat{u} + \beta(\|\hat{x}\|_0 - 1) \Rightarrow \beta \leq \mathbf{r}^T(\hat{x} - \hat{u}).$$

Substituting the decomposition $\hat{x} = g_i(\hat{x}) + \hat{x}[i]e_i$ and the definition of $\hat{u}$, we obtain:

$$\begin{aligned} \beta &\leq \mathbf{r}^T \left(g_i(\hat{x}) + \hat{x}[i]e_i - \frac{t}{\|g_i(\hat{x})\|_D} g_i(\hat{x}) \right) \\ &= \mathbf{r}[i]\hat{x}[i] + \left(1 - \frac{t}{\|g_i(\hat{x})\|_D} \right) \mathbf{r}^T g_i(\hat{x}) \\ &= \mathbf{r}[i]\hat{x}[i] + (\|g_i(\hat{x})\|_D - t) \frac{\mathbf{r}^T g_i(\hat{x})}{\|g_i(\hat{x})\|_D}. \end{aligned}$$

Since $h(0) < 0$, we have $\|g_i(\hat{x})\|_D > t$, so the coefficient $(\|g_i(\hat{x})\|_D - t)$ is positive. By the definition of the dual norm, we have $\frac{\mathbf{r}^T g_i(\hat{x})}{\|g_i(\hat{x})\|_D} \leq H$. Applying this upper bound yields:

$$\beta \leq \mathbf{r}[i]\hat{x}[i] + (\|g_i(\hat{x})\|_D - t)H.$$

Next, using the triangle inequality $\|g_i(\hat{x})\|_D \leq \|\hat{x}\|_D + \|\hat{x}[i]e_i\|_D$ and noting that $\|\hat{x}\|_D = t$, we have:

$$\|g_i(\hat{x})\|_D - t \leq \|\hat{x}[i]e_i\|_D = |\hat{x}[i]| \sqrt{d_i[i]}.$$

Substituting this back into the inequality and using $\mathbf{r}[i]\hat{x}[i] \leq |\mathbf{r}[i]| |\hat{x}[i]|$:

$$\begin{aligned} \beta &\leq |\mathbf{r}[i]| |\hat{x}[i]| + |\hat{x}[i]| \sqrt{d_i[i]} H \\ \beta &\leq |\hat{x}[i]| \left(|\mathbf{r}[i]| + \sqrt{d_i[i]} H \right). \end{aligned}$$

Rearranging for $|\hat{x}[i]|$ gives the desired lower bound:

$$|\hat{x}[i]| \geq \frac{\beta}{H\sqrt{d_i[i]} + |\mathbf{r}[i]|}.$$

We observe that this lower bound is suitable for all three cases above. Finally, if $|\hat{\sigma}| = 1$ then we have closed form solutions for the specific entry $i \in \hat{\sigma}$

$$\hat{x}[i] = \frac{t \text{sign}(\mathbf{r}[i])}{\sqrt{d_i[i]}} \Rightarrow |\hat{x}[i]| = \frac{t}{\sqrt{d_i[i]}}.$$

Combining these two cases will result in

$$i \in \hat{\sigma} \Rightarrow |\hat{x}[i]| \geq \min \left\{ \frac{\beta}{|\mathbf{r}[i]| + H\sqrt{d_i[i]}}, \frac{t}{\sqrt{d_i[i]}} \right\}.$$

□

Emre Can

Proposition 11 *Assuming $\bar{\tau} > 0$ and $H_\omega > \gamma$. The optimal Lagrange multiplier μ^* associated with the constraint in (ZP-RMVP $_\omega$) is given by:*

$$\mu^* = \frac{2\bar{\tau}}{(H_\omega - \gamma)^2}.$$

Proof The problem is convex. We write the Lagrange function using a non-negative multiplier μ as:

$$L(u, \mu) = u^T D_\omega u + \mu (\bar{\tau} - \tau_\omega^T u + \gamma \|u\|_{D_\omega}).$$

The first-order conditions (necessary and sufficient under Slater's condition, provided $H_\omega > \gamma$) give:

$$\nabla_u L(u, \mu) = 2D_\omega u - \mu \tau_\omega + \mu \gamma \frac{D_\omega u}{\|u\|_{D_\omega}} = 0.$$

Let us define $\sigma = \|u\|_{D_\omega}$ and solve for u (assuming $u \neq 0$). Rearranging the terms yields:

$$u = \left(\frac{\mu \sigma}{2\sigma + \mu \gamma} \right) D_\omega^{-1} \tau_\omega.$$

By definition of σ , substituting the expression for u , we have the identity:

$$\sigma^2 = \left(\frac{\mu \sigma}{2\sigma + \mu \gamma} \right)^2 H_\omega^2.$$

Taking the square root (justified by the positivity of σ, μ , and H_ω), we obtain $\sigma = \left(\frac{\mu \sigma}{2\sigma + \mu \gamma} \right) H_\omega$. Rearranging terms, we establish the relation:

$$\mu = \frac{2\sigma}{H_\omega - \gamma} \tag{6}$$

Now, supposing that the constraint is binding, we have the equation $\tau_\omega^T u - \gamma \|u\|_{D_\omega} = \bar{\tau}$. We can express u as $u = \frac{\sigma}{H_\omega} D_\omega^{-1} \tau_\omega$. Substituting this into the linear term $\tau_\omega^T u$ yields:

$$\tau_\omega^T u = \tau_\omega^T \left(\frac{\sigma}{H_\omega} D_\omega^{-1} \tau_\omega \right) = \frac{\sigma}{H_\omega} (\tau_\omega^T D_\omega^{-1} \tau_\omega) = \frac{\sigma}{H_\omega} (H_\omega^2) = \sigma H_\omega.$$

Replacing $\tau_\omega^T u$ with σH_ω and $\|u\|_{D_\omega}$ with σ in the binding constraint equation, we get:

$$\sigma H_\omega - \gamma \sigma = \bar{\tau}.$$

Simplifying via factoring, we obtain:

$$\sigma = \frac{\bar{\tau}}{H_\omega - \gamma},$$

from which we have the condition $H_\omega > \gamma$ for the positivity of σ . Solving for μ from the relation (6) involves substituting this value of σ :

$$\mu = \frac{2\sigma}{H_\omega - \gamma} = \frac{2\left(\frac{\bar{\tau}}{H_\omega - \gamma}\right)}{H_\omega - \gamma} = \frac{2\bar{\tau}}{(H_\omega - \gamma)^2}.$$

Finally, we verify that the constraint must be binding. If the constraint were not binding, the complementary slackness condition would imply $\mu = 0$. Substituting $\mu = 0$ into the first-order condition yields $2D_\omega u = 0$, which implies $u = 0$. However, substituting $u = 0$ into the original constraint gives $0 \geq \bar{\tau}$. Since we assume $\bar{\tau} > 0$, this is a contradiction. Thus, a feasible solution requires the constraint to be active.

Proposition 12 *Assuming $H_\omega > \gamma$, the optimal Lagrange multiplier μ^* associated with the constraint in (ZP-RMVP2) is given by:*

$$\mu^* = H_\omega - \gamma.$$

Proof We write the Lagrange function using a non-negative multiplier λ as:

$$L(y, \mu) = \mathbf{r}_\omega^T y - \bar{\mathbf{r}} - \gamma \|y\|_{D_\omega} + \mu(t - \|y\|_{D_\omega}).$$

Grouping the norm terms, the first-order conditions (necessary and sufficient) give:

$$\nabla_y L(y, \mu) = \mathbf{r}_\omega - (\gamma + \mu) \frac{D_\omega y}{\|y\|_{D_\omega}} = 0.$$

Let us define $\sigma = \|y\|_{D_\omega}$ and solve for y (assuming $y \neq 0$). Rearranging the terms yields:

$$y = \left(\frac{\sigma}{\gamma + \mu} \right) D_\omega^{-1} \mathbf{r}_\omega.$$

By definition of σ , substituting the expression for y , we have the identity:

$$\sigma^2 = \left(\frac{\sigma}{\gamma + \mu} \right)^2 H_\omega^2.$$

Taking the square root (justified by the positivity of σ, μ, γ , and H_ω), we obtain $\sigma = \left(\frac{\sigma}{\gamma + \mu} \right) H_\omega$. Rearranging terms, we establish the relation:

$$\mu = H_\omega - \gamma. \tag{7}$$

Algorithm 1: A Branch-and-Bound Algorithm for (CP-RMVP)

Initialization: Tolerance $\epsilon = 1E - 10$, local and global upper bounds $ub = ub^* = \infty$, a support set denoting possible assets in portfolio $P = \{1, \dots, N\}$, a support set for iteration $S = \emptyset$, a **fixed support** solution $u_P = \xi(P)$ defined in Proposition 1, a lower bound lb for the problem by solving (ZP-RMVP $_{\omega}$) with $\omega = P$, and the componentwise lower bound b for this **fixed support** solution as defined in Proposition ??.

Warm Start: WIP. Enqueue (u_P, ub, lb, P, S) .

while *all nodes are not visited or* $ub^* - lb > \epsilon$ **do**

Extract the tuple t with the highest priority from the queue, $(x, ub, lb, P, S) \leftarrow t$.

if $ub - ub^* < \epsilon$ **then**

 Update the global upper bound $ub^* \leftarrow ub$.

end

if $P \neq \emptyset$ **then**

Branching:

if $|S| \geq 1$ **then**

 Select the asset corresponding to the largest absolute gradient entry:

$j \in \operatorname{argmax}_{i \in P} |\nabla L_i|$.

else

 Select the asset j with the lowest ratio of variance to return:

$j \in \operatorname{argmin}_{i \in P} \operatorname{diag}(D_P)[i]/r_P[i]$.

end

 Branch by creating subproblems with and without each candidate asset:

 – Include Asset: Create a *left node* with support $S^L \leftarrow S \cup \{j\}$.

 – Exclude Asset: $P \leftarrow P \setminus \{j\}$ and create a *right node* with support , $S^R \leftarrow P \cup S$

Bounding Right Node:

if $|S^R| \geq 2$ **then**

 Calculate a lower bound $lb^R = \xi(S^R)^T D_{S^R} \xi(S^R) + \beta |S|$.

 Enqueue (x, lb^R, ub, P, S) .

end

Bounding Left Node:

if $|S^L| \geq 1$ **then**

 Calculate an upper bound $ub^L = \xi(S^L)^T D_{S^L} \xi(S^L) + \beta |S^L|$.

 Enqueue $(\xi(S^L), lb + \beta, ub^L, P, S^L)$.

else

 Enqueue $(x, lb + \beta, ub, P, S^L)$.

end

else

 Continue to the next iteration.

end

end

return: ub^* and its corresponding solution.

Notes about experiments

$\gamma = 0.1$ drops almost all assets

SP500 $\beta = 0.01$ returns one sparse solution

DowJones $\beta = 0.1, 0.05, 0.01$ returns one sparse solution. $\beta = 0.001$ returns 1-2-3 sparse. $\beta = 0.005$ returns 1-6 sparse. $\beta = 0.0001$ returns 2-10 sparse.

FTSE100 $\beta = 0.1, 0.05, 0.01, 0.001$ returns one sparse solution. $\beta = 0.0001$ returns 3-4 sparse solution.

NASDAQ100 (n=70) $\beta = 0.1, 0.05, 0.01$ returns one sparse solution. $\beta = 0.001$ returns 1-3 sparse solution.

ItalianBonds (n=11) $\beta = 0.001, 0.0005$ returns 1 sparse. $\beta = 0.0001$ returns 1-2 sparse. $\beta = 0.00005$ returns 1-3 sparse. $\beta = 0.00001$ returns 2-3 sparse.

Buse

References